Lipid Chemistry

- Lipids are a heterogenous group of water insoluble organic molecules, that can be extracted from tissues by non polar solvents.
- Make up 18–25% of body mass in lean adults
- Formed of long-chain hydrocarbon groups, also contain oxygen, phosphorus, nitrogen and sulfur.
- Ratio of hydrogen to oxygen is not 2:1,they are hydrophobic.

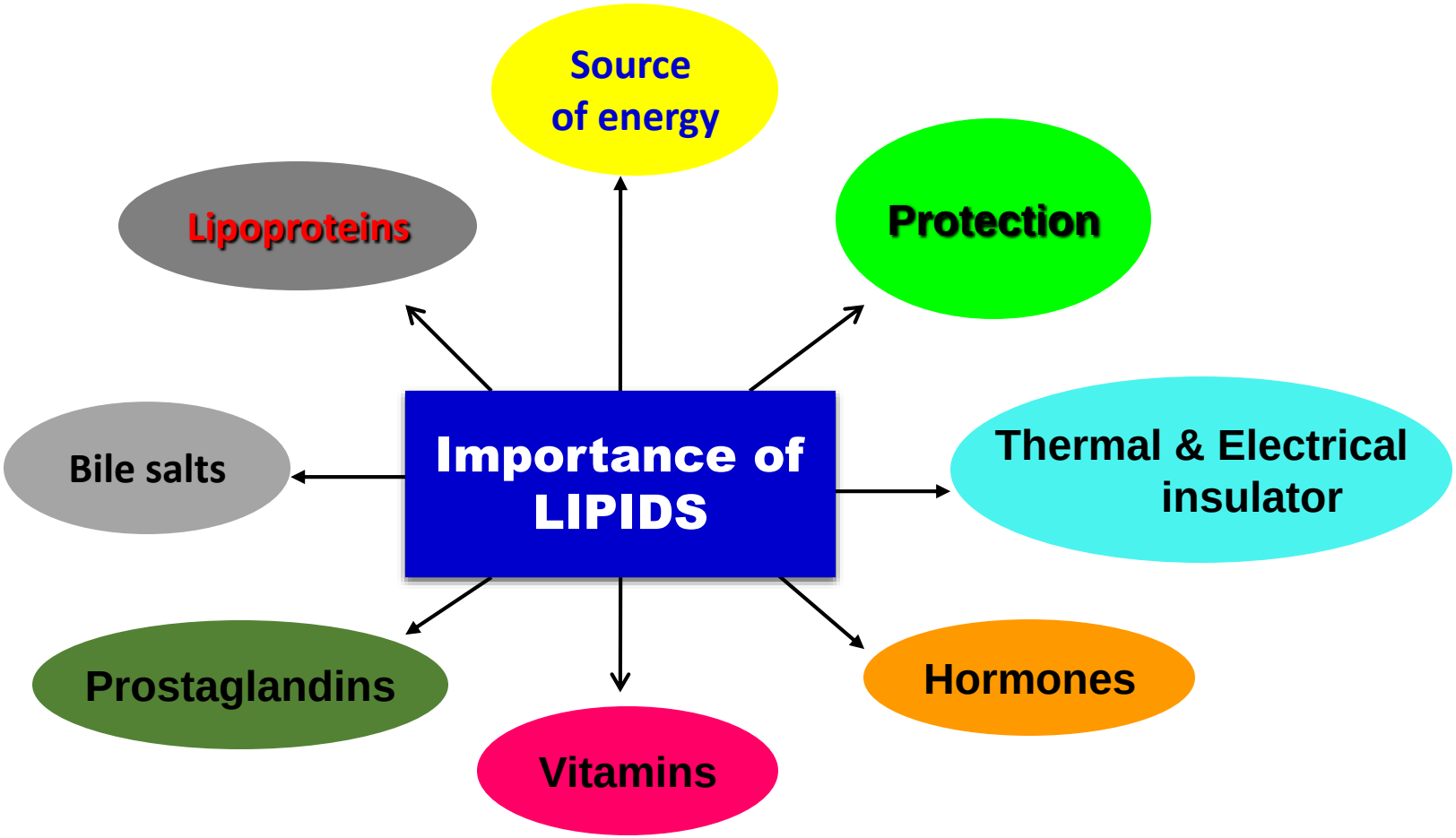
- Related more by their physical than by their chemical properties.
- include fats, oils, steroids, waxes, and related compounds
- Bloor`s Criteria
- Relatively insoluble in water
- Soluble in nonpolar solvents such as ether and chloroform.
- Actually or potentially related to Fatty acids
- Utilized by living organisms

Biomedical Importance of Lipids

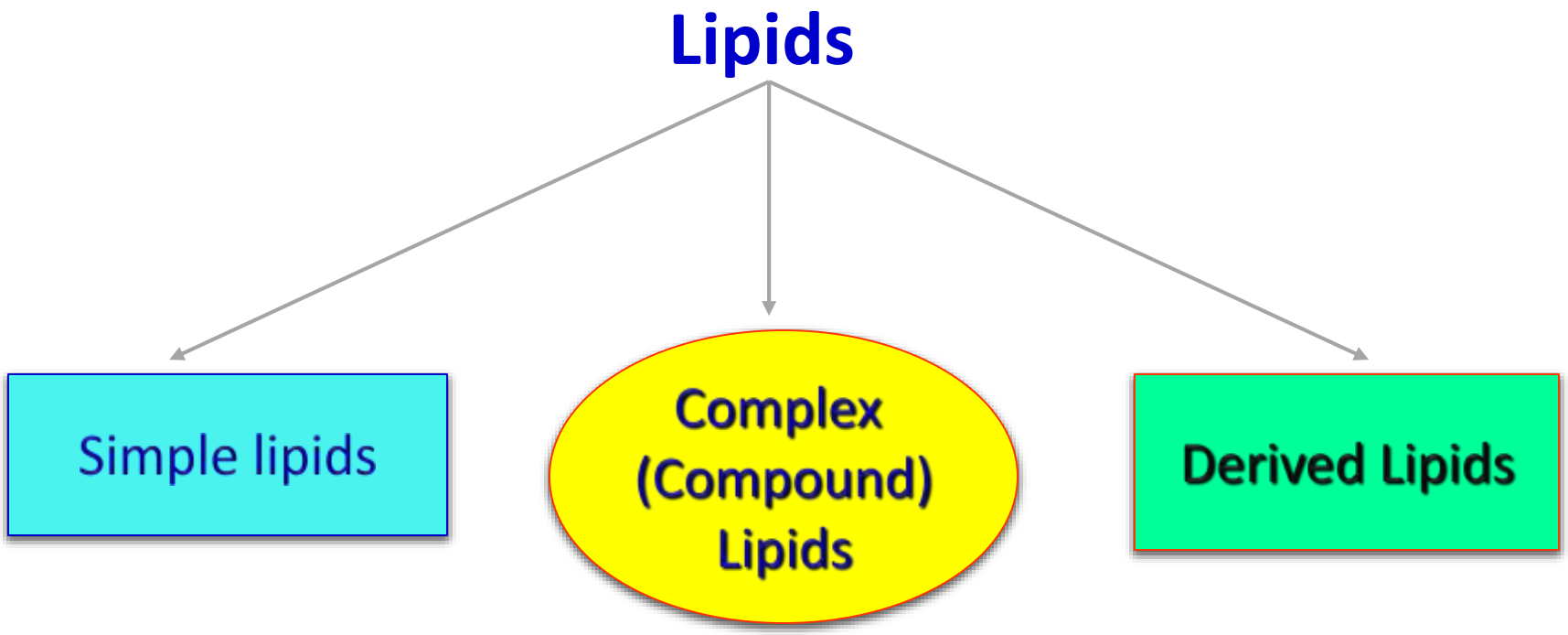
- Important source of energy (oxidation of 1g of fat yields 9.3 Kilocalories)
- Protective coating and around certain organs to keep them in position
- Thermal insulator.
- Electrical insulator.
- Prostaglandins, some hormones, and some vitamins are lipids in nature.

- Bile salts are derived lipids that help the digestion and absorption of lipids.
- Lipoproteins (complexes of lipids and proteins) are important cellular constituents, and help the transport of lipids in plasma.
- Important in understanding many areas of interest like obesity and atherosclerosis.

Biomedical Importance of Lipids



Chemical Classification of Lipids



- **Simple lipids:** Esters of fatty acids with various alcohols.

a. Neutral Fats: Esters of fatty acids with glycerol. uncharged. Eg.- triolein, tripalmitin.

Oils - Fats in the liquid state. Eg.- corn, groundnut oil.

b. **Waxes:**

True Waxes: Esters of fatty acids (C14 – 36) with higher M.wt monohydric long chain alcohols (C16 - 30)

Eg. Bees wax, lanolin, spermaceti oil.

Used for lotions, ointments

Other waxes: cholesterol ester, Vit.A ester, Vit.D ester.

- Complex Lipids:
- Esters of fatty acids with alcohol + containing additional groups (prosthetic group).
- Phospholipids: Fatty acids ,alcohol and a phosphate group . nitrogen containing bases and other substituents.
  E.g. Glycerophospholipids the alcohol is glycerol
  Sphingophospholipids the alcohol is sphingosine.
- Glycolipids (glycosphingolipids): fatty acid, sphingosine and carbohydrate with nitrogen base.
- Other complex lipids: Lipoproteins, sulfolipids and aminolipids.

- Derived lipids: Derived from simple lipids or complex lipids on hydrolysis.
- Fatty acids, glycerol.
- Monoacyl and Diacylglycerol.
- Steroids- lipids with cyclopentanoperhydrophenanthrene (steroid) ring.
- E.g. – cholesterol and its derivatives- bile acids, hormones, Ergosterol.
- Eicosanoids – Arachidonic acid derivatives- PG,$TXA_{2,}$

- Terpenes: Isoprene units – 5 Carbon compounds

  E.g. Carotenes, vitamins A,E,K and Dolichol,CO-Q, Squalene.

- Others: Fatty aldehydes, ketone bodies, hydrocarbons and other alcohols.

  Neutral lipids - uncharged .

  E.g. acylglycerols, cholesterol, and cholesteryl esters.



Obesity

Diabetes Mellitus



Fatty acids

- Simplest form of lipids
- Aliphatic monocarboxylic organic acids with hydrocarbon side chain
- FA are included in the group of derived lipids
- Most common component of lipids in the body
- Free FA are formed only during metabolism
- General formula R-COOH.
- R → Alkyl / hydrocarbon chain

  COOH→ Carboxyl end
- Occurs mainly as ESTERS in natural fats and oils.

Even and Odd carbon fatty acids

- Fatty acids that occur in natural lipids are of even carbons (usually 14C – 20C)
- This is due to biosynthesis of fatty acids mainly occurs with the sequential addition of 2 carbon units.
- Palmitic acid (16C) and Stearic acid (18C) are most common.
- Propionic acid (3C) and valeric acid (5C) are common odd chain fatty acids.

Nomenclature

- Systematic name
- It is based on hydrocarbon from which it is derived.
- After the name of the parent hydrocarbon
- With suffix – anoic acid for saturated fatty acid,
- – enoic acid for unsaturated fatty acid.

8 7 6 5 4 3 2 1

CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-**COOH**

Hydrocarbon chain Carboxyl group

Octane + acid = Octanoic acid

- Common name – Caprylic acid

Numbering

- The Carbon atoms are numbered from COOH group as 1.
- The carbons adjacent to this are 2,3,4 etc or α, β, γ
- Carboxyl group carbon is C_1, next carbon atom is C_2 / α-carbon, next is β and so on,
- Last carbon atom or CH_3 group, ω / n carbon.

Numbering of fatty acids

ω

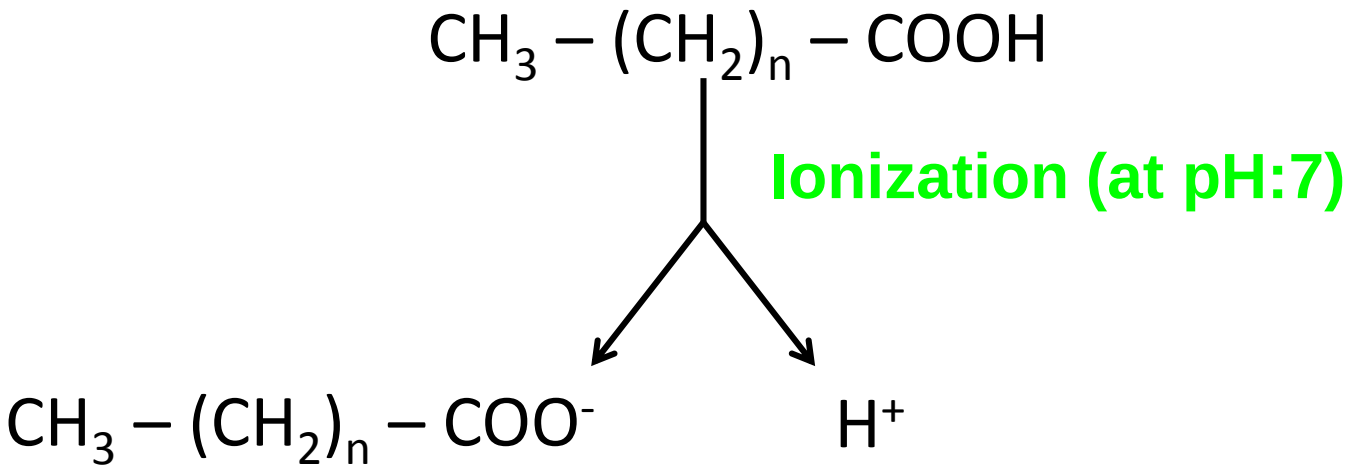
$CH_3 – CH_2 - CH_2 - COOH$ **Butyric acid**

4 3 2 1 *(Arabic numbers)*

γ β α *(Greek alphabetical numbers)*

1 2 3 4 *(Omega numbers)*

General Formula of Fatty Acids



General Structure of a Fatty Acid

-Fatty acids are amphipathic molecules **composed of**

a hydrophilic (polar, ionized) head (formed by the carboxyl group) and a hydrophobic (non-polar, non-ionized) tail (formed by the hydrocarbon chain).

-The degree of solubility of a fatty acid depends on the length of the hydrocarbon chain.



Non-polar

polar

H_2O-insoluble

H_2O-soluble

Classification of Fatty Acids

1. According to chain length

   short, medium and long

2. According to degree of saturation

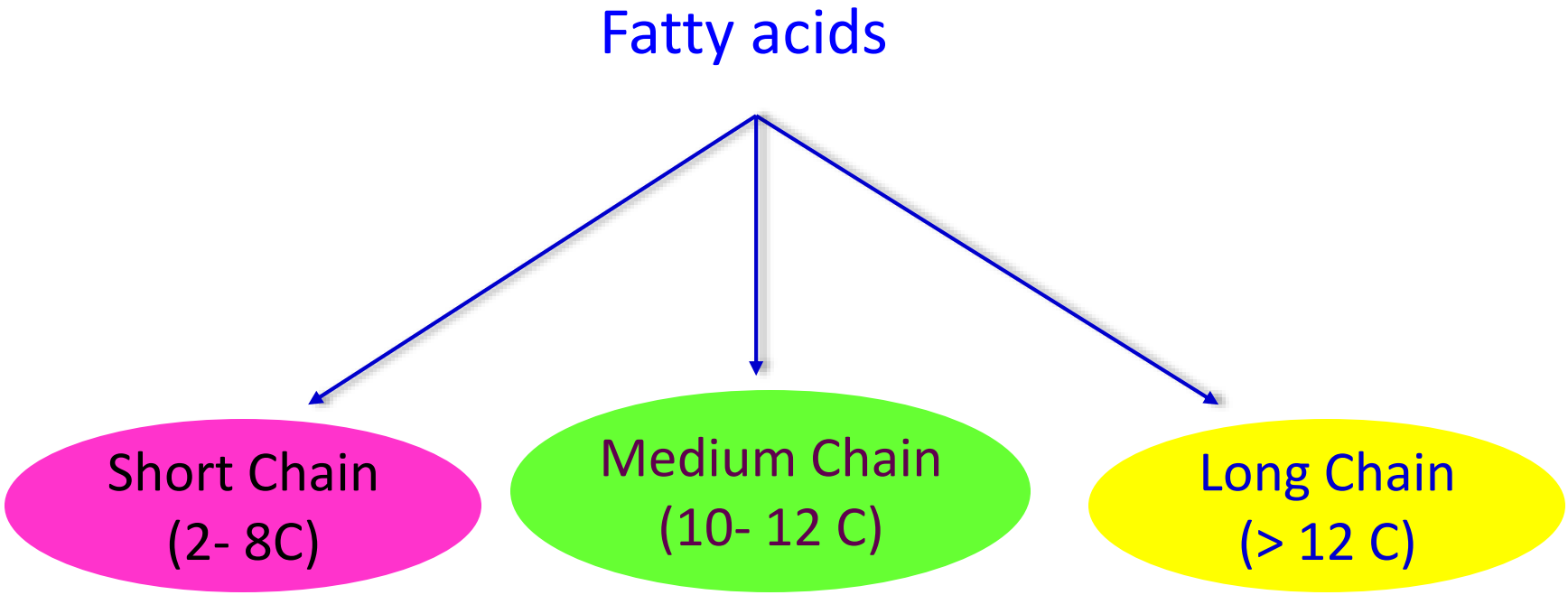
   saturated & unsaturated

3. According to Biological value

   essential & non-essential

4. Aliphatic, branched and cyclic

According to Chain Length



Classification of Fatty Acids

A. According to chain length

1. Low or short chain fatty acids

- Contain 8 carbon atoms or less (from 2-8)
- Acetic acid (2 C): CH_3-COOH
- Butyric (4 C): CH_3-CH_2-CH_2-COOH

2. Medium-chain fatty acids

- Contain from 10-12 carbon atoms.

3. High or Long chain fatty acids

- Contain more than 12 carbon atoms
- Palmitic (16 C): CH_3-$(CH2)_{14}$-COOH
- Stearic (18 C): CH_3-$(CH2)_{16}$-COOH
- Lignoceric (24 C): CH_3-$(CH2)_{22}$-COOH

Classification of fatty acids according to degree of saturation

- Saturated Fatty Acids
- **They do not contain double bonds**

- **Palmitic (16 C)**

CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-COOH

[CH3-$(CH2)_{14}$-COOH]

- **Stearic acid (18 C)**

CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-COOH

[CH3-$(CH2)_{16}$-COOH]

Unsaturated Fatty Acids

- They contain one or more double bonds
- They are named by adding suffix “enoic”
- Unsaturated fatty acids exhibit the geometrical isomerism at the double bonds.
- All naturally occurring fatty acids have the cis-configuration.

- e.g. Oleic acid (18: 1; 9) **ω**9

CH_3- $(CH_2)_7$ - CH = CH - $(CH_2)_7$ – COOH

Classification of Unsaturated Fatty Acids

- Monounsaturated fatty acids (monoethenoid or monoenoic)
- They contain one double bond

  e.g. Oleic acid (18: 1; 9) ω9

  CH_3-$(CH_2)_7$-CH = CH-$(CH_2)_7$-COOH
- Polyunsaturated fatty acids (polyethenoid or polyenoic)
- They contain more than one double bond

Polyunsaturated Fatty Acids

1. Dienoic fatty acids (contain 2 double bonds)

e.g. linoleic acid (18: 2 ; 9,12) ω6.

12 9

$CH_3 - (CH2)_4 - CH = CH - CH_2 - CH = CH - (CH2)_7 - COOH$

2. Trienoic fatty acids (contain 3 double bonds)

e.g. α-Linolenic (18: 3; 9,12, 15) ω3

15 12 9

$CH_3 - CH2 - CH = CH - CH_2 - CH = CH - CH_2 - CH = CH - (CH2)_7 - COOH$

$CH_3 - (CH_2 - CH = CH)_3 - (CH_2)_7 - COOH$

3. Tetraenoic fatty acids (contain 4 double bonds)

e.g. Arachidonic (20: 4; 5, 8, 11, 14) ω 6

$CH_3 - (CH_2)_3 - (CH_2 - CH = CH)_4 - (CH_2)_3 - COOH$

Classification of FA according to Biological Value

- Essential Fatty Acids:
- The FA that cannot be synthesized by the body, should be supplied through diet known as essential fatty acids.
- They are polyunsaturated fatty acids, linoleic acid and linolenic acid.

- NOTE:
- Mammals can synthesize saturated fatty acids (FAs) and monounsaturated FAs, but they are unable to synthesize FAs containing more than one double bond because they lack the enzyme system that is responsible for introduction of more than one double bond.
- Arachidonic acid is synthesized in the body from linoleic acid, so it is a non-essential fatty acid.
- It is also found in animal fats and peanut oil.
- Arachidonic acid becomes essential if its precursor, linoleic acid, is missing in the diet.

Functions of essential fatty acids

- Structural elements of tissues:
- PUFA occur in higher concentration in lipids in associated with structural elements of tissues.
- Structural elements of gonads:
- Lipids of gonads also contain a high concentration of PUFA, which suggests importance of these compounds in the reproductive function.
- Synthesis of prostaglandins:
- Prostaglandins are synthesized from arachidonic acid by cyclooxygenase enzyme system.
- Leukotriens are conjugated trienes formed from arachidonic acid in leucocytes by the lipoxygenase pathway.

- Structural element of mitochondrial membrane:
- Defeciency of EFA causes swelling of mitochondrial membrane and reduction in efficiency of oxidative phosphorylation.
- Serum level of cholesterol:
- Fats with high content of PUFA tends to lower serum level of cholesterol.
- Effect on clotting time:
- Prolongation of clotting time is noted in ingestion of fats rich in EFA.
- Role of EFA in fatty liver: Deficiency produces fatty liver.

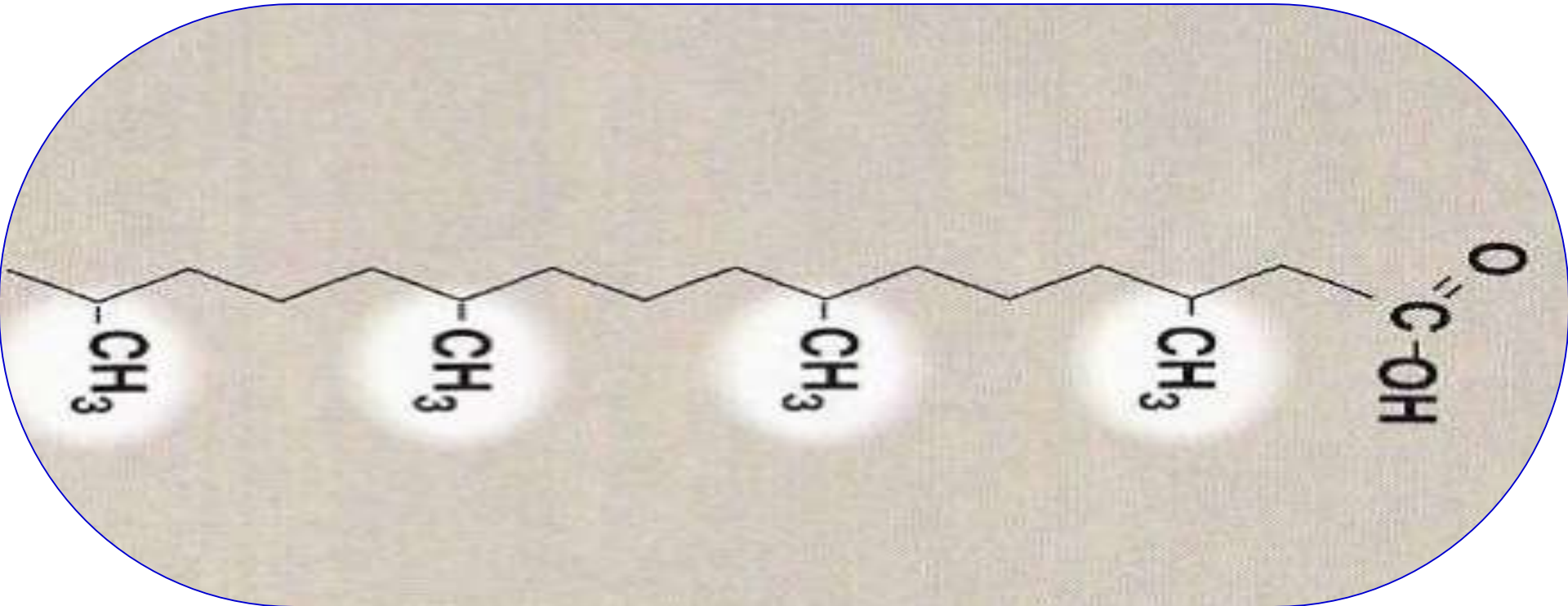
- Role in vision:
- Docosahexenoic acid is most aboundent polyeneoic fatty acid present in retinal photoreceptor membranes.
- Decosahexenoic acid is formed from dietary linolinic acid.
- It enhences the electrical response of the photoreceptors to illumination.
- Hence linolenic acid is necessary for optimal vision.

Deficiency of EFA

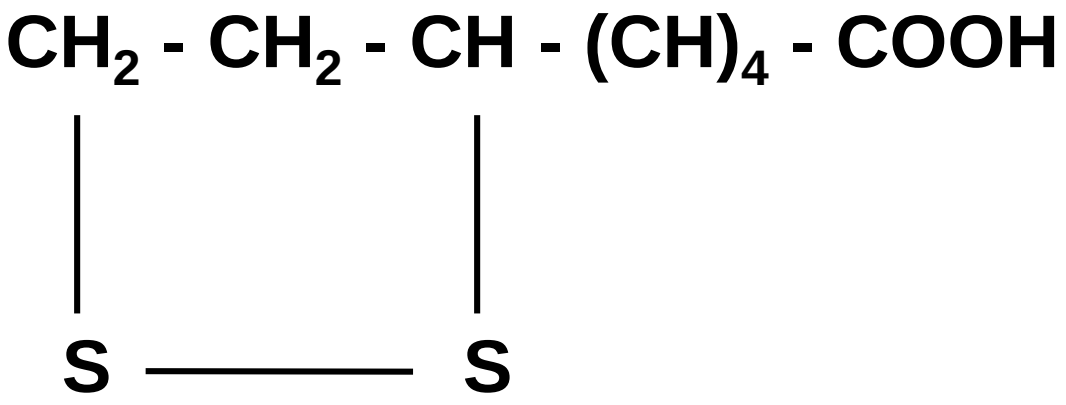
- The deficiency of EFA results in phrynoderma or toad skin, characterized by the presence of horney eruptions on the posterior and lateral parts of limbs, on the back and buttocks, loss of hair and poor wound healing.
- Non Essential Fatty Acids:
- They can be synthesized by mammals, so it is not essential to take them in diet.
- They include saturated fatty acids and mono-unsaturated fatty acids.

Branched Fatty Acids

- Phytanic acid (3,7,11,15 tetra methyl palmitic acid)
- It is present in milk lipids and animal fat.



Cyclic Fatty Acids



- Lipoic acid:
- It is a hydrogen carrier and coenzyme in oxidative decarboxylation of α-ketoacids e.g. pyruvate and α-ketoglutarate dehydrogenase complexes.

Eicosanoids
(Prostaglandins, Thromboxanes and Leukotrienes)

Chemistry:

1. Twenty carbon compounds.
2. Derived from arachidonic acid.
3. Include Prostaglandins (PGs), thromboxanes (TXs) and leukotrienes (LTs).
4. PGs have several classes (A-H) of which PGs A,E and F are the major classes in humans.

Triacylglycerol

- All the commercially important fats and oils of animal and plant origin consist exclusively of the simple lipid class triacylglycerols
- Esters of fatty acid with the trihydric alcohol-glycerol.
- Glycerol with one molecule of fatty acid is called monoacylglycerol.
- Two molecule of fatty acid is diacylglycerol.
- Three fatty acids is triglyceride.
- Fats as stored fuel:
- TAGs are the most aboundent group of lipids that primarily function as fuel reserves of animals.

- Fats primarily occur in adipose tissue:
- Adipocytes of adipose tissue –predominantly found in the subcutaneous layer and in the abdominal cavity –are specialized for storage of TAGs.
- Simple TAG contain the same type of fatty acid residue at all the three carbons
- Mixed triacylglycerols are more common.
- In general, FA attached to C1 is saturated, that attached to C2 is unsaturated while that on C3 can be either.

Properties of TAGs

- Hydrolysis:
- TAGs undergo enzymatic hydrolysis to finally liberate free fatty acid and glycerol.
- The process of hydrolysis, catalyzed by *lipases* is important for digestion of fat in GIT and fat mobilization from the adipose tissue.
- Saponification:
  The hydrolysis of TAG by alkali to produce glycerol and soaps is known as saponification.

Triacylglycerol + 3NaOH → Glycerol + 3 R-COONa (soaps)

Rancidity

- Rancidity is the term used to represent the deterioration of fats and oils resulting in the unpleasant taste.
- Fats containing unsaturated fatty acids are more susceptible to rancidity.
- Rancidity occurs when fats and oils are exposed to air, moisture, light, bacteria etc.
- Hydrolytic rancidity occurs due to partial hydrolysis of TAG by bacterial enzymes.
- Oxidative rancidity is due to oxidation of unsaturated fatty acids.
- Results in the formation of unpleasant products such as dicarboxylic acids, aldehydes, ketones etc.
- Rancid fats and oils are unsuitable for human consumption.

Antioxidants

- The substances which can prevent the occurrence of oxidative rancidity are known as antioxidants.
- Trace amounts of antioxidants such as tocopherols (vitamin E), hydroquinone, gallic acid and α-naphthol are added to commercial preparations of fats and oils to prevent rancidity.

Lipid peroxidation

- In the living cells, lipids undergo oxidation to produce peroxides and free radicals which can damage the tissue.
- The free radicals are believed to cause inflammatory diseases, cancer, atherosclerosis etc.
- It is fortunate that the cells possess antioxidants such as vitamin E, urate and superoxide dismutase (SOD) to prevent in vivo lipid peroxidation.

Tests to check purity of fats and oils

- Iodine number:
- It is defined as the grams (number) of iodine absorbed by 100 g of fat or oil.
- Iodine number is useful to know the relative unsaturation of fats, and is directly proportional to the content of unsaturated fatty acids.
- Thus lower is the iodine number, less is the degree of unsaturation.

Iodine number

Fat/oil	Iodine number
Coconut oil	7 - 10
Butter	25 - 28
Palm oil	45 - 55
Olive oil	80 - 85
Groundnut oil	85 - 100
Cottonseed oil	100 - 110
Sunflower oil	125 - 135

Saponification number

- Defined as the mg (number) of KOH required to hydrolyse (saponify) one gram of fat or oil.
- Saponification number is a measure of the average molecular size of the fatty acids present.
- The value is higher for fats containing short chain fatty acids

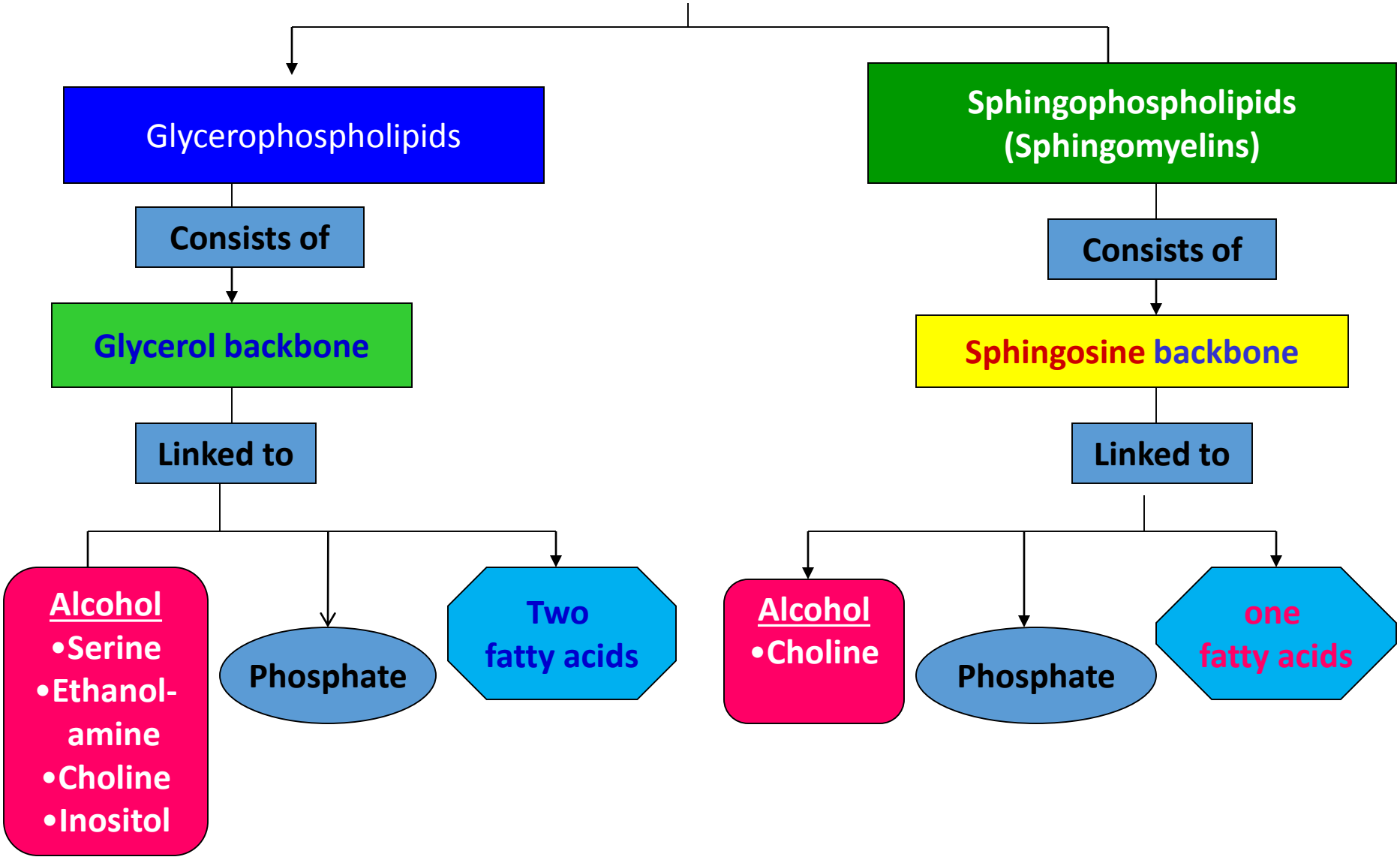
Human fat	195 – 200
Butter	230 – 240
Coconut oil	250 - 260

- **Reichert – Meissl (RM) number:**
- It is defined as the number of ml 0.1N NaOH required to completely neutralize the soluble volatile fatty acids distilled from 5 g of fat.
- RM number is useful in testing the purity of butter.
- **Acid number:**
- It is defined as the number of mg of KOH required to completely neutralize free fatty acids present in one gram of fat or oil.
- Oils with increased acid number are unsafe for human consumption.
- Chemical or bacterial contamination – yield free fatty acids.

Phospholipids

- They are polar (ionic) compounds composed of an alcohol that is attached by a posphodiester bridge to either diacyl glycerol or sphingosine.
- Classification of Phospholipids
- Phospholipids are classified into 2 classes:
- Glycerophospholipids:
- They have glycerol as a backbone.
- Sphingophospholipids (Sphingomyelins):
- They have sphingosine as a backbone.

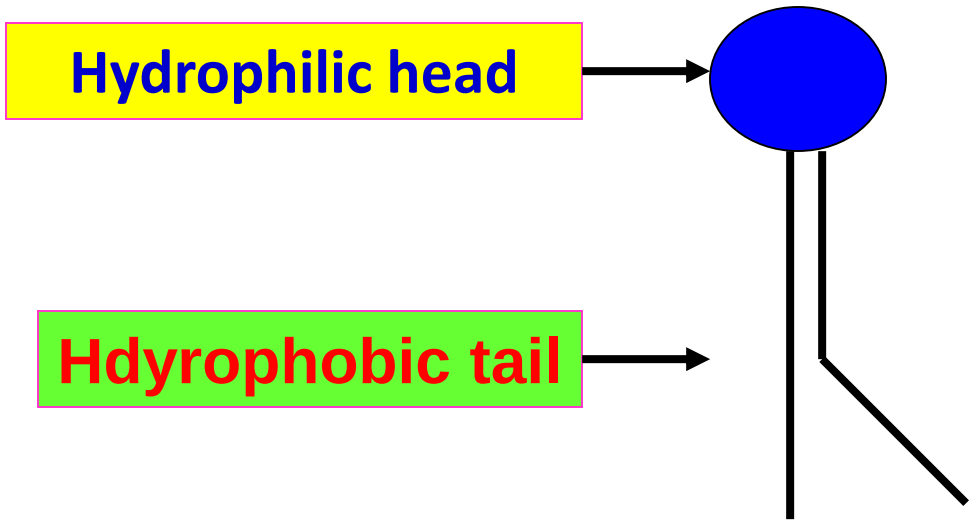
Structure of Phospholipids



Properties of Phospholipids

- They are amphipathic molecules i.e. each molecule has a hydrophilic (polar) head (formed of the phosphate group and the alcohol group) and a hydrophobic (non-polar) tail (formed of glycerol or sphingosine and the hydrocarbon chains of the fatty acids).

Phospholipids are Amphipathic Molecules



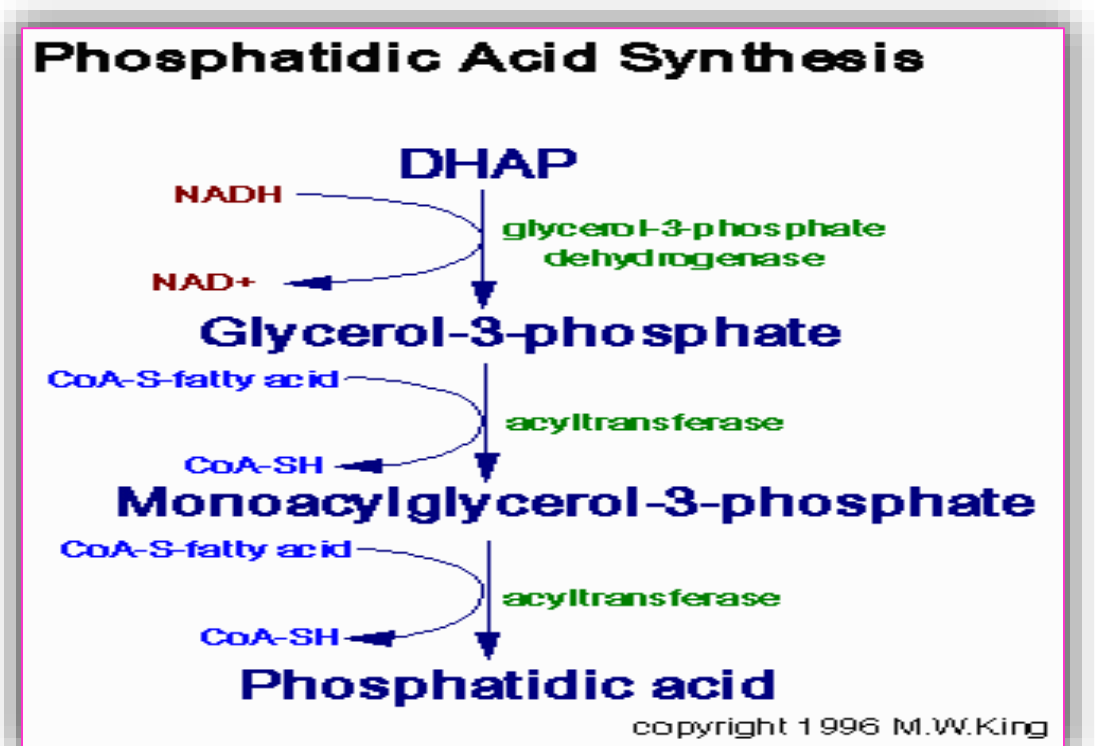
Phospholipid molecule

Members of Glycerophospholipids

1. Phosphatidic acid (parent phospholipids).
2. Phosphatidyl choline (Lecithin).
3. Phosphatidyl ethanolamine (cephalin).
4. Phosphatidyl serine.
5. Phosphatidyl inositol.

Phosphatidic acid (parent phospholipids)

- This is the simplest phospholipid.
- It does not occur in good concentration in the tissues.
- Phosphatidic acid is an intermediate in synthesis of triacylglycerols and phospholipids.

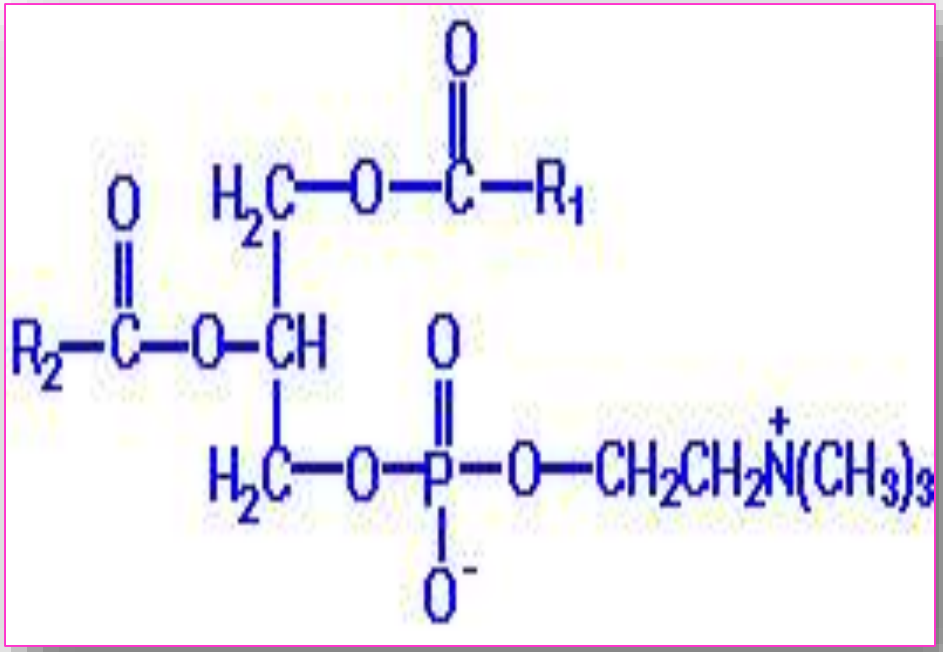


Lecithin

- Lecithin:
- Alcohol + fatty acid + phosphoric acid + choline.
- Present in brain, nervous tissue, sperm and egg yolk.
- Are surface-active agent and help in emulsification of fats.
- Dipalmitoyl lecithin is a lung surfactant (lowers surface tension) prevents the collapse of lung alveoli.

- Absence of dipalmitoyl lecithin in premature infants may produce respiratory distress syndrome (RDS) or hyaline membrane disease.

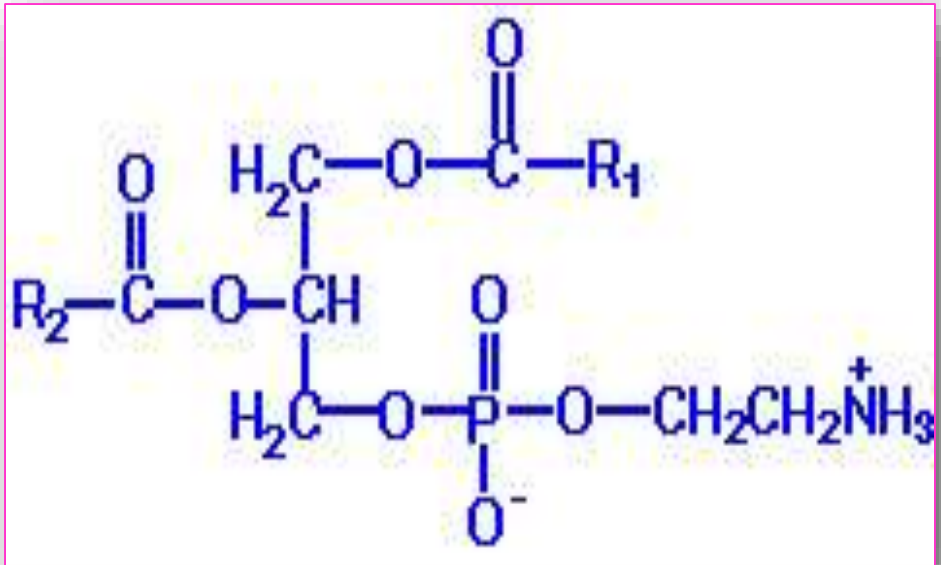
Structure of lecithin



Cephalins

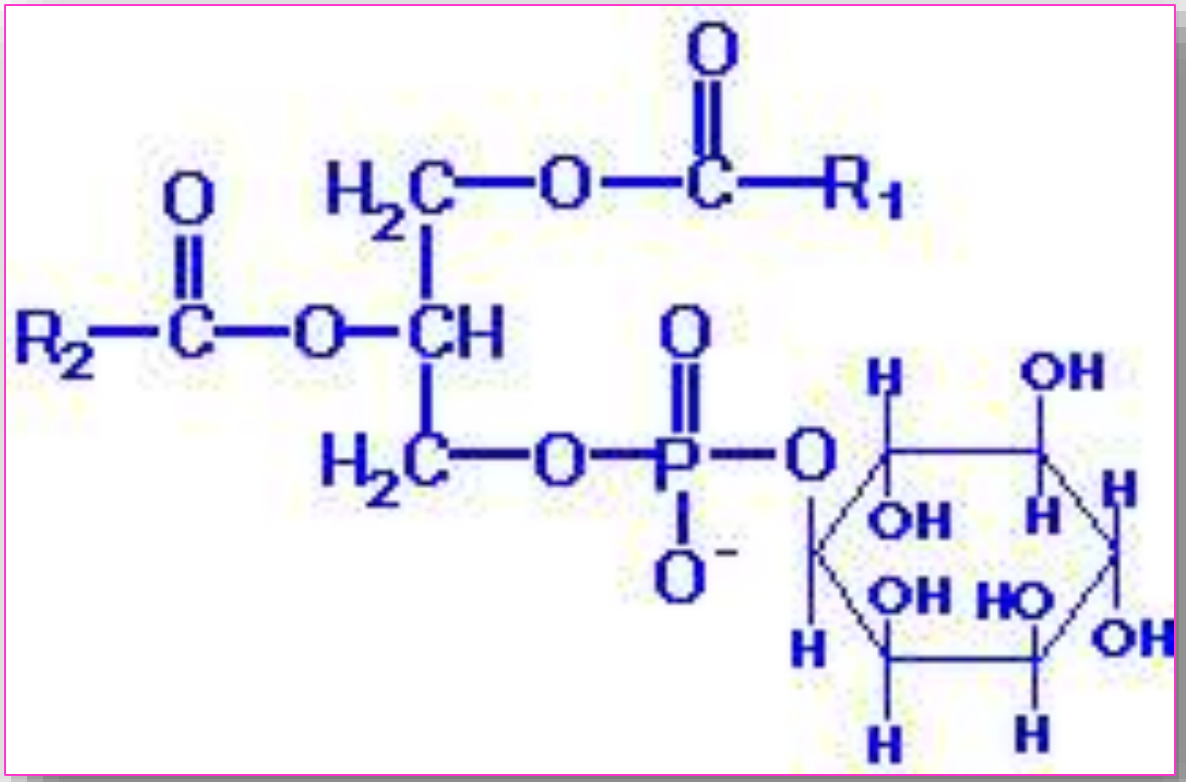
- Cephalins:
- Alcohol+ fatty acid+ phosphoric acid + ethanolamine or serine as a nitrogenous base instead of choline present in lecithin.
- Present in brain, erythrocytes and many other tissues.

Structure of cephalins
(Phosphatidyl ethanolamine)



Phosphatidyl inositol

- Phosphatidyl Inositol:
- Phosphatidic acid + inositol (alcohol) instead of a nitrogenous base.
- They are important component of cell membrane.
- The action of certain hormones (e.g. oxytocin, vasopressin) is mediated through phosphatidyl inositol (PI).
- In response to hormonal action, PI is cleaved to diacyl glycerol (DAG) and inositol triphosphate (IP3).
- Both these compounds act as second messenger for hormonal action.

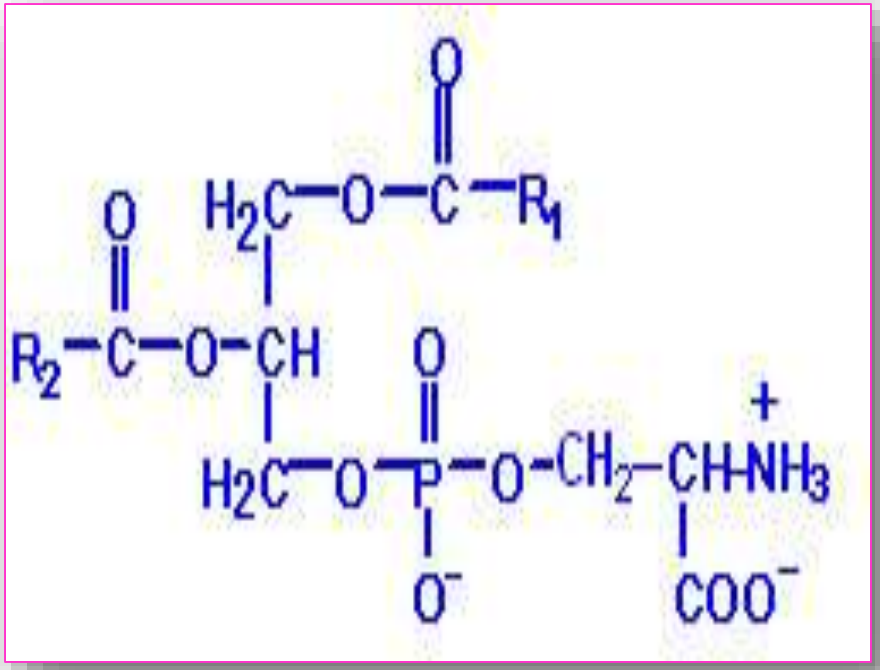


Structure of Phosphatidyl inositol

Phosphatidyl serine

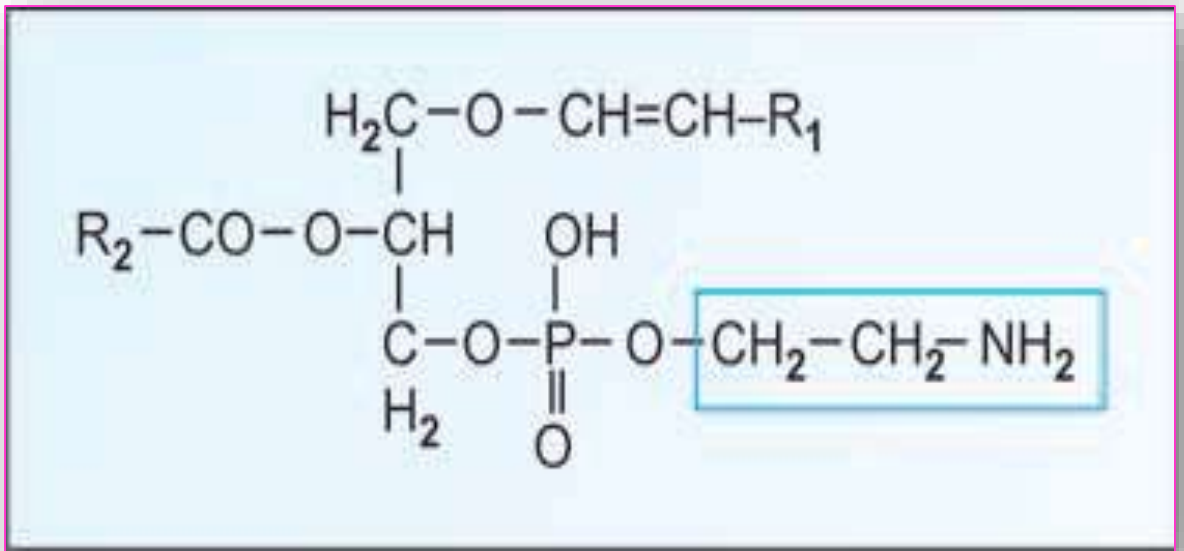
- Phosphatidyl serine:
- The amino acid serine is present in this group of glycerophospholipids.
- Present in brain, nervous tissue and small amount in other tissues.

Structure of Phosphatidyl serine



Plasmalogen

- They differ from lecithin or cephalin in α1 position of glycerol where the fatty acid is replaced by a long chain unsaturated aliphatic aldehyde such as palmitic or stearic aldehyde.
- Plasmalogens are present in large quantities in the skeletal muscle, cardiac muscle and in semen.

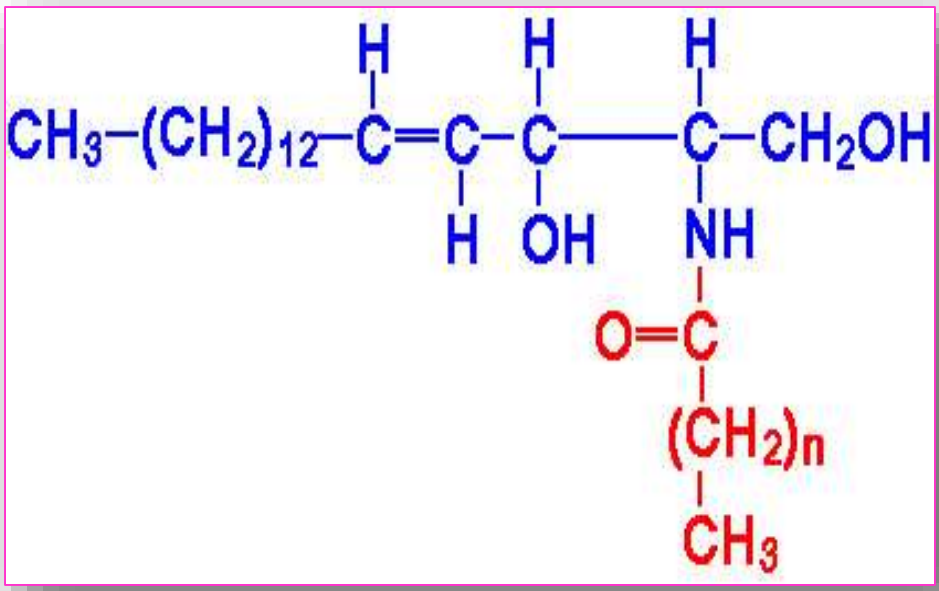


Cardiolipin

- It is diphosphatidyl glycerol.
- Contains two molecules of phosphatidic acid held by glycerol.
- Present in the inner mitochondrial membrane and has antigenic properties.

Ceramide

- Ceramide:
- Formed by the esterification of sphingosine with a fatty acid of high molecular weight.
- Found in white matter of brain and nerves.
- Ceramide is common for all glycolipid and sphingomyelin.

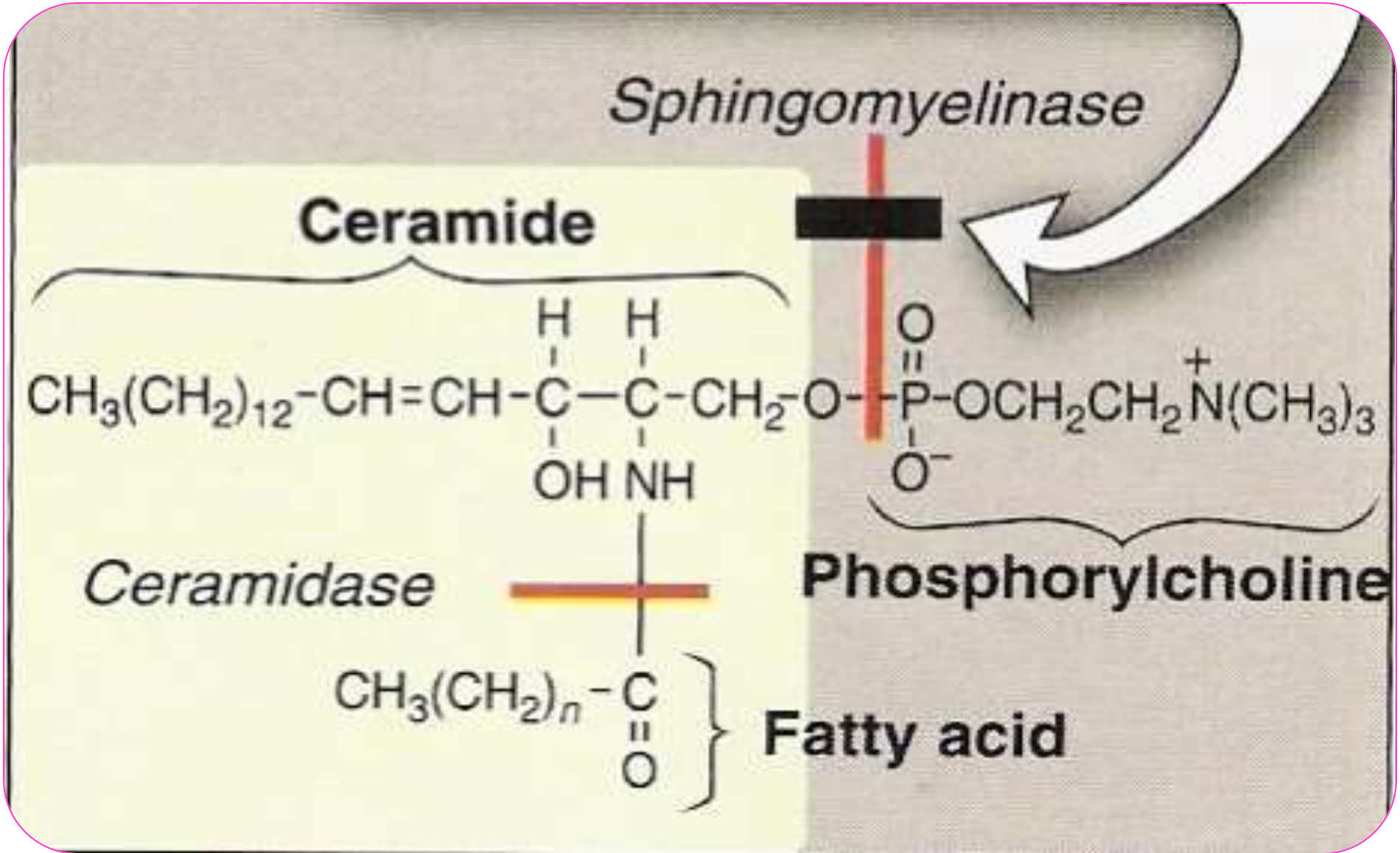


Structure of ceramide

Sphingomyelin

- This is a sphingophospholipid.
- Does not contain glycerol but a unsaturated amino alcohol, i.e. sphingosine.
- Contain a molecule of choline, phosphoric acid and FA
- Sphingomyelin makes up a large part of the myelin sheath.
- These are also present in brain, lungs, nerve and other tissues.
- Deposition of sphingomyelin in liver, lymph nodes, bone marrow and central nervous system results in Neimman-Pick disease due to deficiency of sphingomyelinase enzyme in these tissues.

Sphingophospholipids (Sphingomyelins)



Functions of phospholipids

- Phospholipids participate in the lipoprotein complexes which are thought to constitute the matrix of cell walls and membranes, the myelin sheath.
- Role in enzyme action:
- Certain enzymes require tightly bound phospholipids for their action, e.g.mitochondrial enzyme system involved in oxidative phosphorylation.
- Role in blood coagulation:
- Phospholipids play an essential part in the blood coagulation process required at Conversion of prothrombin to thrombin by active factor X.

- Possibly also in the activation of factor VIII by the activated factor IX.
- Role in lipid absorption in intestine:
- Lecithin lowers the surface tension of water and aids in emulsification of lipids, a prerequisite in digestion and absorption of lipids from GIT.
- Role in transport of lipids from intestine:
- Exogenous TG is carried as lipoprotein complex, chylomicrons, in which PL takes an active part.

- Role in transport of lipids from liver:
- Endogenous TG is carried from liver to various tissues as lipoprotein complex Pre –β- lipoprotein complex (VLDL).
- PL is required for the formation of lipoprotein complex.
- Role in electron transport:
- Phospholipids help to couple oxidation with phosphorylation and maintain electron transport enzymes in active conformation and proper relative positions.
- Lipotropic action of lecithin:
  Choline acts as a lipotropic agent as it can prevent formation of fatty liver

- Membrane phospholipids are source of Arachidonic acid.
- Insulation:
- Phospholipids of myelin sheaths provide the insulation around the nerve fibres.
- Cofactor:
  Phospholipidsare required as a cofactor for the activity of the enzyme lipoprotein lipase and triacylglycerol lipase.
- Dipalmityl lecithin is an important phosphatidylcholine found in lungs.
- It is a surface active agent and prevents the adherence of inner surface of the lungs due to surface tension.

- *Respiratory distress syndrome* in infants is a disorder characterized by the absence of dipalmityl lecithin.
- Lecithin-Sphingomyelin Ratio(L/S ratio):
- L/S ratio in amniotic fluid has been used for the evaluation of fetal maturity.
- Prior to 34 weeks gestation, the amniotic fluid lecithin and concentrations are approximately equal.
- After this time, there is a marked increase in lecithin and L/S ratio greater than 5
- A L/S ratio of > 2 or > 5 indicate adequate fetal lung maturity.
- Delivery of a premature low weight fetus, with L/S ratio 1, indicate that the infant will probably develop respiratory distress or hyaline membrane disease.

Glycolipids

- They contain fatty acid, sphingosine (amino alcohol), carbohydrate or carbohydrate derivative.
  e.g. Cerebroside, ganglioside
- Cerebrosides:
- Contains fatty acid, sphingosine and a sugar (usually galactose).
- Present in white matter of brain and myelin sheath of nerves.
- Increased in Gaucher's disease in tissues like reticuloendothelial cells of spleen, liver, lymph node and bone.

Gangliosides

- Contains N-acetylneuramininc acid (sialic acid), fatty acid, sphingosine and three molecules of hexoses (glucose or galactose).
- Found in grey matter of the brain
- In Tay Sach's disease ganglioside level increases.

Lipoproteins

- The different types of lipoproteins are
- Chylomicron:
- Transports dietary triglyceride and cholesterol esters from intestine to peripheral tissues and liver
- Very low density lipoprotein (VLDL):
- Transports endogenous triglyceride from liver to extrahepatic tissues.
- Low density lipoprotein (LDL):
- Transports cholesterol from liver to extrahepatic tissues.

- High density lipoprotein (HDL):
- Transports cholesterol from extrahepatic tissues to the liver in an esterified form.
- Since lipids are water insoluble they are present in the blood in the form of lipoproteins which are water-soluble.
- Composed of triglyceride and cholesterol ester core surrounded by a shell of proteins (also called as Apo proteins), phospholipid and free cholesterol.
- Separated by ultracentrifugation into four distinct groups based on their density (d) and by electrophoresis

Steroids

- Found in association with lipids.
- Having special ring, cyclopentanoperhydrophenanthrene nucleus

  Eg. Steroid hormone, bile acid, vitamin D.

Cholesterol

- One of the important steroids present in the body.
- Has 27 carbon, an -OH group, a double bond, two methyl groups at C10 and C13 and a side chain at C17.
- Precursor of various compounds such as vitamin D3, bile acids and adrenocortical and sex hormones.
- Cholesterol is widely distributed in all cells of the body but nervous tissue is rich in cholesterol.
- Steroids containing one or more -OH groups are known as Sterols

- Normal fasting serum cholesterol level is 150-200 mg/dl.
- It is synthesized in our body using acetyl CoA as precursor Cholesterol exists in free and ester form.
- Cholesterol gets esterified through esterase enzymes.
- Excess is harmful to body, it gets deposited in the intima of the arteries producing atherosclerosis.
- This can narrow the lumen of blood vessel impeding blood flow, which cause thrombosis.

Functions of cholesterol

- Cholesterol if maintained in normal level it has number of good effects.
- It is a precursor for the synthesis of bile acids in liver.
- The steroid hormone in adrenal cortex and sex hormones in gonads are mainly synthesized from cholesterol.
- Cholesterol form 7 dehydrocholesterol in skin, it is converted to vitamin D3 by UV rays.

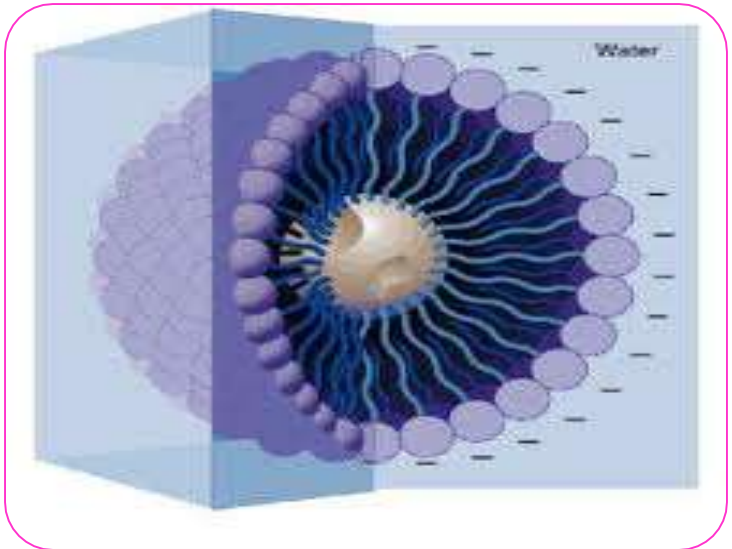
- It is a poor conductor of heat and hence acts as an insulator
- Cholesterol is abundant in brain and nervous tissue where it functions as an insulating covering for structure, which generates and transmits electrical impulse.

Amphipathic lipids

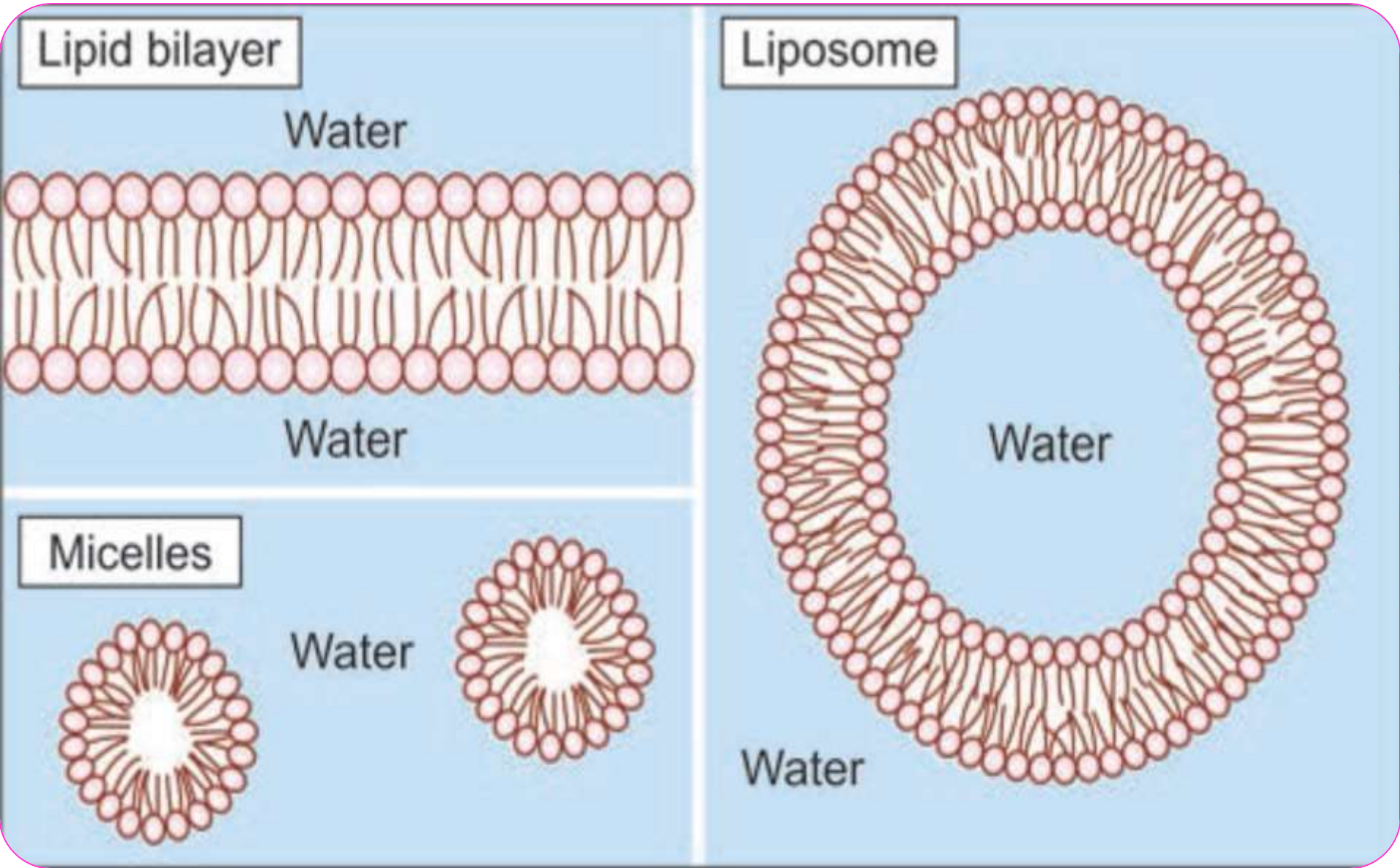
- Lipids are insoluble in water.
- This is primarily due to the presence of hydrocarbon groups.
- Some of the lipids possess polar or hydrophilic groups which tend to be soluble in water.
- *Molecules which contain both hydrophobic and hydrophilic groups are known as amphipathic.*
- Examples of amphipathic lipids:
  Fatty acids, phospholipids, sphingolipids, bile salts and cholesterol (to some extent) are amphipathic in nature.

Orientation amphipathic lipids

- When the amphipathic lipids are mixed in water (aqueous phase), the polar groups (heads) orient themselves towards aqueous phase while the non – polar (tails) orient toward the opposite direction.
- This leads to the formation of micelles.
- Micelle formation, facilitated by bile salts is very important for lipid digestion and absorption.



Structure of Mixed Micelles



Reference Books

- **Harper's Illustrated Review: BIOCHEMISTRY**
- **Lippincott's Illustrated Reviews: BIOCHEMISTRY**
- Test Book of Medical Biochemistry – MN Chatterjea

